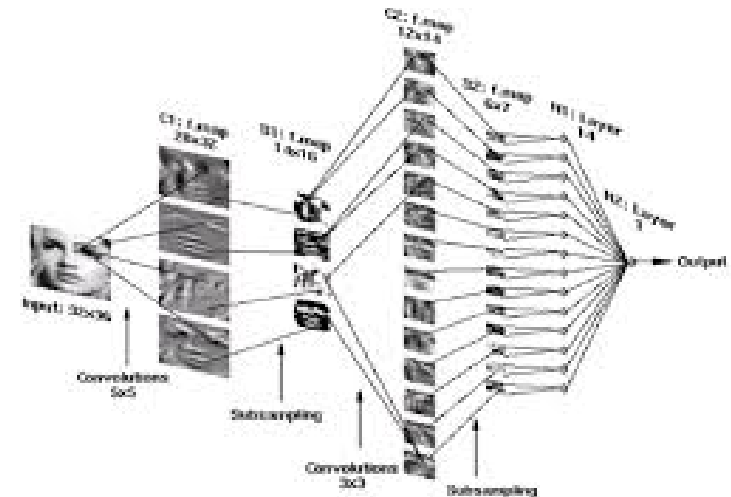


Outline

1. Why are you doing this course?
2. What we assume that you know



Orientation

- This course addresses three cohorts of students:
 - ★ Third year Artificial Intelligence Students
 - ★ MSc in Machine Learning Students
 - ★ Volunteers
- It is dual coded (AICE3001/AICE6002), with subtle difference in expectations reflected in slightly different evaluation
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- This course is aimed at building up trust in your understanding of how machine learning works

Why Mathematics?

- Learning machines are notoriously difficult to understand
- Mathematics allows us to reason rigorously
- Posing question mathematically and seeking mathematical solutions provide allow us to build a strong intuition
- Mathematics also allow us to describe the world precisely
- It also provides an arsenal of tools that turn out to be hugely useful

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Cracking the Code

- Mathematics is a language that takes time to master
- For some it is easy, for others it will be a struggle
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Starting from Different Places

- You all start from different places
- Some of you may have a degree in mathematics
- For others it may have been your least favourite subject
- I invite strong mathematicians to become even stronger
- For students starting from a weak base, judge yourselves by your progress, not by comparing yourself to others

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The Journey

- We are going to visit lots of aspects of mathematics that have proved fruitful in machine learning
- Some of this will be very concrete, while in other cases by look at a higher level of abstraction we can generalise methods across different application areas
- We will cover, calculus, vector spaces, convexity, optimisation, entropy, probability, statistics and much else besides
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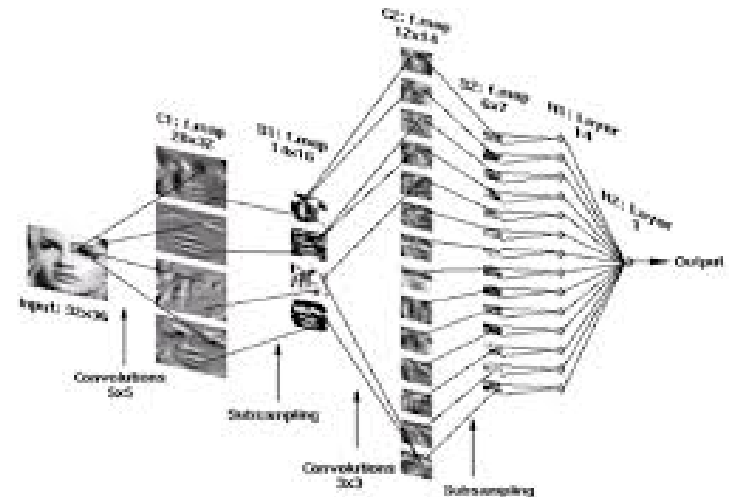
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What we Assume you Know

- You should all have A-level mathematics and have done some mathematics at Uni
- You should know what a function is, e.g. $f(x) = x^2 + 3$
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Vectors

- Ordinary vectors (what mathematicians call elements of $\mathbb{R}^n$) are much used in machine learning and should be familiar
- We will use bold symbols to represent **column vectors**, e.g.

$$\boldsymbol{x} = \begin{pmatrix} 1.2 \\ 2.3 \\ 4.7 \end{pmatrix}$$

- We denote **row vectors** as the **transpose** of column vectors
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Matrices

- Matrices are blocks of number that define linear mappings between vectors, e.g. $\mathbf{M} \in \mathbb{R}^{2 \times 3}$

$$\mathbf{M} = \begin{pmatrix} 1.1 & -2.1 & 6.2 \\ 9.3 & 3.1 & -4.4 \end{pmatrix}$$

- An $m \times n$ matrix (i.e. $\mathbf{M} \in \mathbb{R}^{m \times n}$) has m rows and n columns
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Composition of Mappings

- Matrices, $\mathbf{A}$, $\mathbf{B}$, etc. represent mappings between spaces
- Provided the matrices are the correct size we can compose these mappings, e.g. if $\mathbf{A} \in \mathbb{R}^{\ell \times m}$ and $\mathbf{B} \in \mathbb{R}^{m \times n}$ then $\mathbf{AB}$ represents the effect of
 - ★ first applying $\mathbf{B}$ to vectors in $\mathbb{R}^n$ to obtain vectors in $\mathbb{R}^m$
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Matrix Multiplication

- The combined mapping $\mathbf{AB}$ can be represented by an $\ell \times n$ matrix $\mathbf{C}$ which we can compute using matrix multiplication (rows times columns)

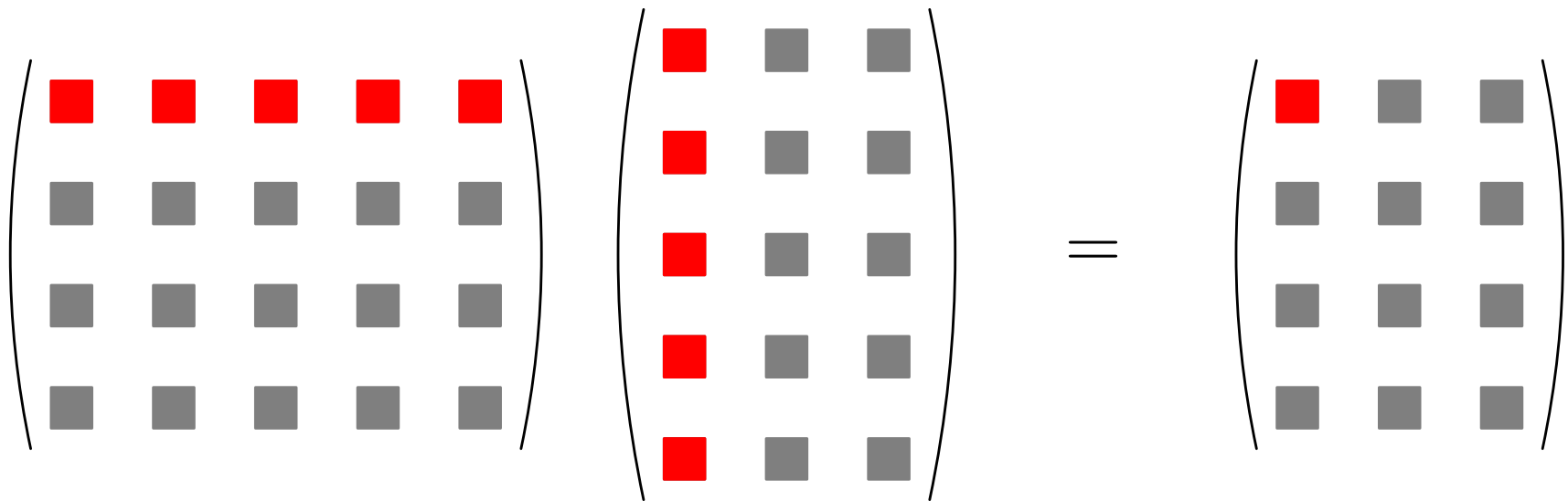
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The diagram illustrates matrix multiplication using a visual representation of matrices as grids of squares. The first matrix, representing $\mathbf{A}$, is a 5x5 grid with 5 rows and 5 columns. The second matrix, representing $\mathbf{B}$, is a 5x3 grid with 5 rows and 3 columns. An equals sign follows, leading to the resulting matrix $\mathbf{C}$, which is a 5x3 grid with 5 rows and 3 columns. This visualizes the operation $\mathbf{AB} = \mathbf{C}$.

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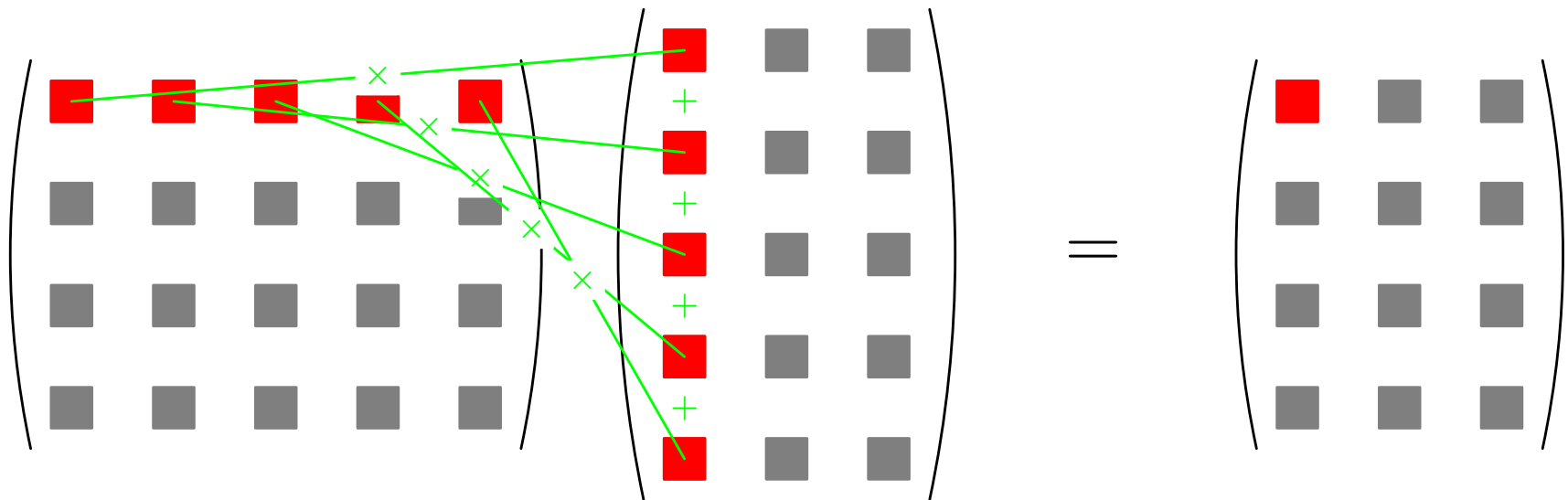
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The diagram illustrates matrix multiplication using a grid of colored squares. The first matrix (A) is 4 rows by 5 columns, with all squares colored red. The second matrix (B) is 5 rows by 3 columns, with all squares colored gray. An equals sign follows, leading to the resulting matrix (C), which is 4 rows by 3 columns, with all squares colored gray. This represents the equation $\mathbf{AB} = \mathbf{C}$.

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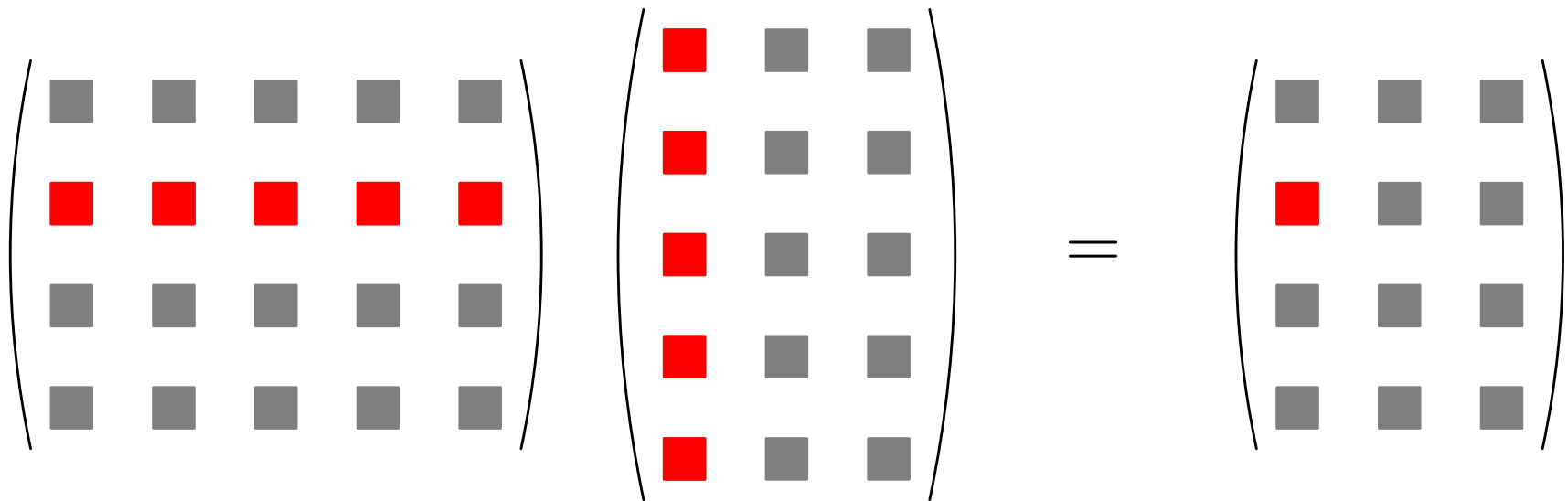
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Gray	Gray	Gray	Gray	Gray	=	Gray	Gray	Gray
Red	Red	Red	Red	Red		Gray	Red	Gray
Gray	Gray	Gray	Gray	Gray		Gray	Gray	Gray
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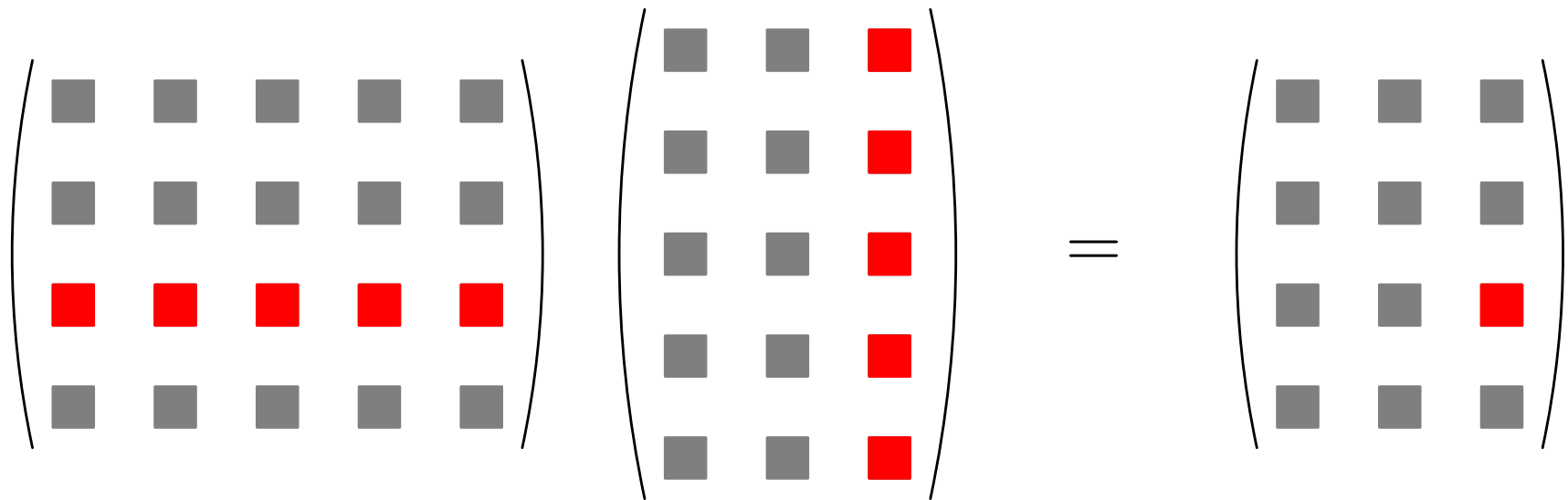
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Matrix A (5x5)	Matrix B (5x3)	Result Matrix C (5x3)
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Gray	Gray	Gray
Red	Red	Red
Gray	Gray	Gray
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$$\begin{pmatrix} \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{red} & \text{red} & \text{red} & \text{red} & \text{red} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \end{pmatrix} \begin{pmatrix} \text{gray} & \text{gray} & \text{red} \\ \text{gray} & \text{gray} & \text{red} \\ \text{gray} & \text{gray} & \text{red} \\ \text{gray} & \text{gray} & \text{red} \\ \text{gray} & \text{gray} & \text{red} \end{pmatrix} = \begin{pmatrix} \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{red} \\ \text{gray} & \text{gray} & \text{gray} \end{pmatrix}$$

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The diagram illustrates matrix multiplication using a grid of squares. The first matrix is a 4x5 grid with 4 rows and 5 columns. The bottom row is red, and all other squares are gray. The second matrix is a 5x3 grid with 5 rows and 3 columns. The first column is red, and all other squares are gray. An equals sign follows, and the resulting matrix is a 4x3 grid with 4 rows and 3 columns. The bottom-left square is red, and all other squares are gray.

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The diagram illustrates matrix multiplication using a grid of squares. The first matrix is a 4x5 grid with 4 rows and 5 columns. The bottom row is highlighted in red, while the other three rows are gray. The second matrix is a 5x3 grid with 5 rows and 3 columns. The middle column is highlighted in red, while the other two columns are gray. An equals sign follows, leading to the resulting 4x3 matrix. This matrix has 4 rows and 3 columns. The bottom-middle square (row 4, column 2) is highlighted in red, while the other squares are gray.

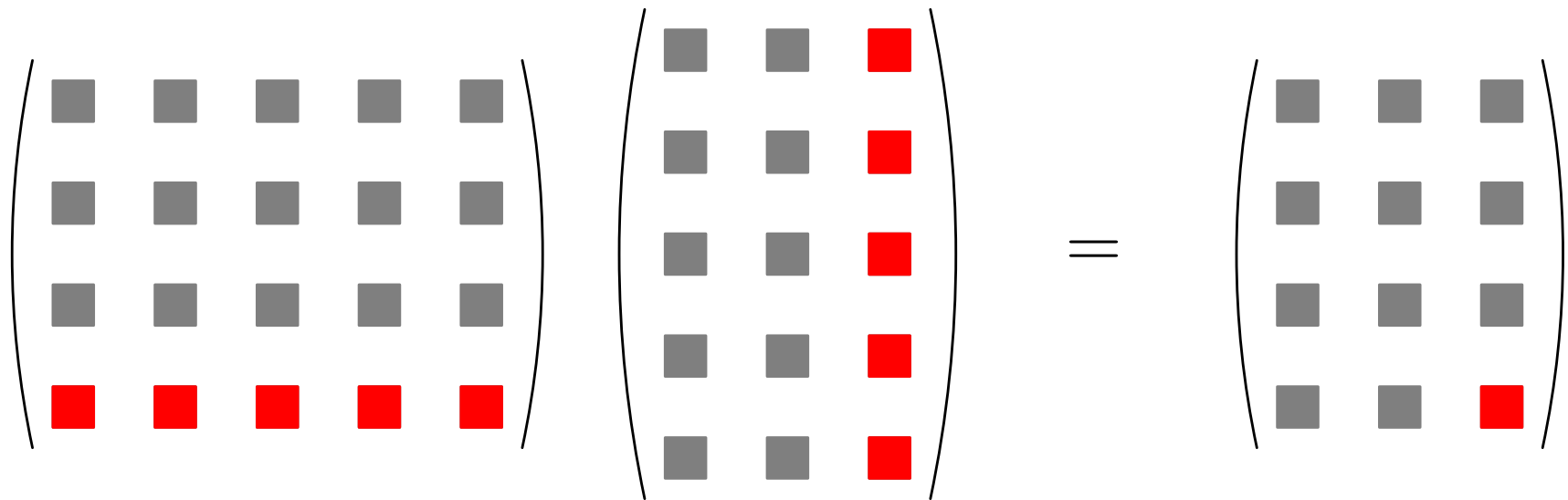
Matrix Multiplication

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The diagram illustrates matrix multiplication using a visual representation of matrices as grids of squares. The first matrix (A) is 4 rows by 5 columns, with the bottom row highlighted in red. The second matrix (B) is 5 rows by 3 columns, with the third column highlighted in red. An equals sign follows, and the resulting matrix (C) is 4 rows by 3 columns, with the third column highlighted in red. This represents the operation $\mathbf{AB} = \mathbf{C}$.

Matrix Multiplication

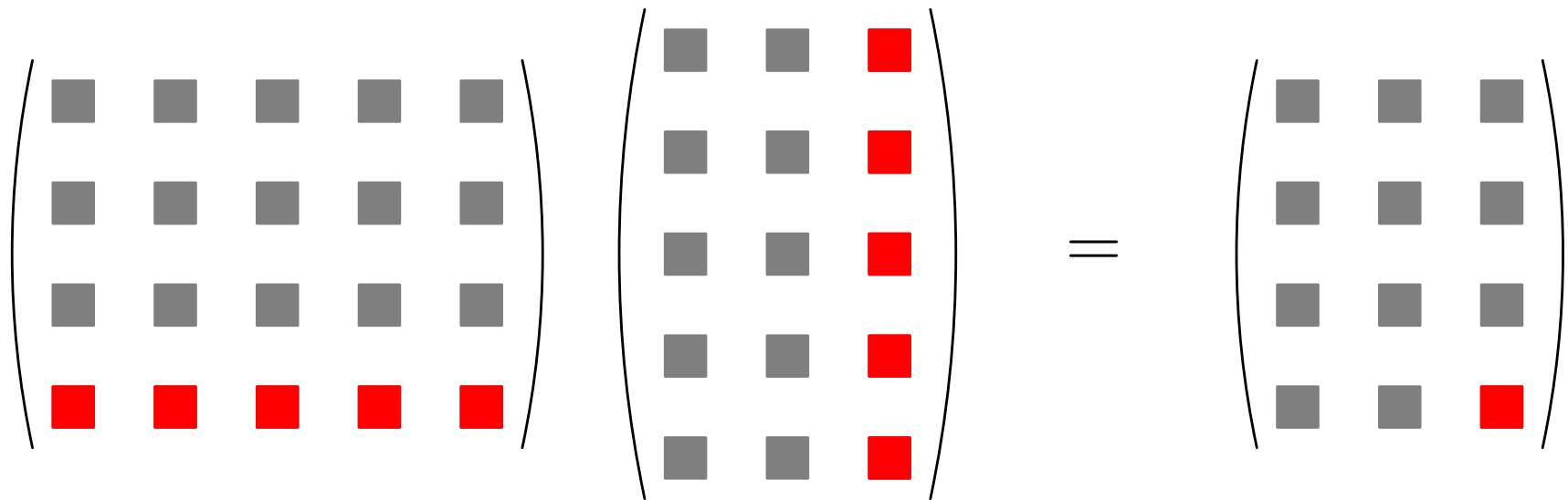
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- Matrix multiplication is associative: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$

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- Matrix multiplication is associative: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$
- But in general not commutative $\mathbf{AB} \neq \mathbf{BA}$

Transpose

- We often encounter transposes of a matrix $\mathbf{M}^T$ where we have exchanged the rows and columns: $[\mathbf{M}^T]_{ij} = [\mathbf{M}]_{ji}$
- Clear $(\mathbf{M}^T)^T = \mathbf{M}$
- It is simple question of book keeping to show that

$$(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$$

- For a symmetric matrix $\mathbf{M} = \mathbf{M}^T$
- Since 1×1 matrices are always symmetric

$$\mathbf{a}^T \mathbf{b} = (\mathbf{a}^T \mathbf{b})^T = \mathbf{b}^T \mathbf{a}$$

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- Clear $(\mathbf{M}^\top)^\top = \mathbf{M}$
- It is simple question of book keeping to show that

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$$

- For a symmetric matrix $\mathbf{M} = \mathbf{M}^\top$
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$$\begin{aligned} \mathbf{a}^\top \mathbf{b} &= (\mathbf{a}^\top \mathbf{b})^\top = \mathbf{b}^\top \mathbf{a} \\ \mathbf{a}^\top \mathbf{M} \mathbf{b} &= \mathbf{b}^\top \mathbf{M}^\top \mathbf{a} \end{aligned}$$

Matrix-Vector Multiplication

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The diagram illustrates matrix-vector multiplication as matrix multiplication. On the left, a 4x5 matrix is represented by a large left parenthesis followed by a 4x5 grid of 20 small gray squares, followed by a large right parenthesis. This is multiplied by a 5x1 column vector, represented by a large left parenthesis followed by a vertical column of 5 small gray squares, followed by a large right parenthesis. An equals sign follows, and then a 4x1 column vector is shown, represented by a large left parenthesis followed by a vertical column of 4 small gray squares, followed by a large right parenthesis.

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The diagram illustrates matrix-vector multiplication as matrix multiplication. It shows a 4x5 matrix (represented by a grid of 20 squares, with the top row highlighted in red) multiplied by a 5x1 column vector (represented by a single column of 5 red squares). This is followed by an equals sign and a 4x1 column vector (represented by a single column of 4 squares, with the top square highlighted in red). The entire expression is enclosed in large parentheses.

$$\begin{pmatrix} \text{4x5 matrix} \end{pmatrix} \begin{pmatrix} \text{5x1 column vector} \end{pmatrix} = \begin{pmatrix} \text{4x1 column vector} \end{pmatrix}$$

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The diagram illustrates matrix-vector multiplication as matrix multiplication. It shows a 5x5 matrix (with 1 red square in the second row) multiplied by a 5x1 column vector (all red squares), resulting in a 5x1 column vector (with 1 red square in the second row).

$$\begin{pmatrix} \text{grey} & \text{grey} & \text{grey} & \text{grey} & \text{grey} \\ \text{red} & \text{red} & \text{red} & \text{red} & \text{red} \\ \text{grey} & \text{grey} & \text{grey} & \text{grey} & \text{grey} \\ \text{grey} & \text{grey} & \text{grey} & \text{grey} & \text{grey} \\ \text{grey} & \text{grey} & \text{grey} & \text{grey} & \text{grey} \end{pmatrix} \begin{pmatrix} \text{red} \\ \text{red} \\ \text{red} \\ \text{red} \\ \text{red} \end{pmatrix} = \begin{pmatrix} \text{grey} \\ \text{red} \\ \text{grey} \\ \text{grey} \\ \text{grey} \end{pmatrix}$$

Matrix-Vector Multiplication

- We can treat a column vector as a $n \times 1$ matrix
- Then matrix-vector multiplication can be treated the same as matrix multiplication

The diagram illustrates matrix-vector multiplication as matrix multiplication. It shows a 5x5 matrix (represented by a grid of squares, with the third row highlighted in red) multiplied by a 5x1 column vector (represented by a single column of squares, with all squares highlighted in red). The result is a 5x1 column vector (represented by a single column of squares, with the third square highlighted in red). The equation is shown as:

$$\begin{pmatrix} \text{gray} & \text{gray} & \text{red} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{red} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{red} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{red} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{red} & \text{gray} & \text{gray} \end{pmatrix} \begin{pmatrix} \text{red} \\ \text{red} \\ \text{red} \\ \text{red} \\ \text{red} \end{pmatrix} = \begin{pmatrix} \text{gray} \\ \text{gray} \\ \text{red} \\ \text{gray} \\ \text{gray} \end{pmatrix}$$

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The diagram illustrates matrix-vector multiplication as matrix multiplication. It shows a 5x5 matrix (represented by a grid of 25 squares, with the bottom row highlighted in red) multiplied by a 5x1 column vector (represented by a single column of 5 red squares). The result is a 5x1 column vector (represented by a single column of 5 squares, with the bottom square highlighted in red). The entire equation is enclosed in large parentheses.

$$\begin{pmatrix} \text{5x5 Matrix} \end{pmatrix} \begin{pmatrix} \text{5x1 Column Vector} \end{pmatrix} = \begin{pmatrix} \text{5x1 Column Vector} \end{pmatrix}$$

Inner Products

- The dot or inner-product between two vectors with the same number of elements, can be written $a^T b$

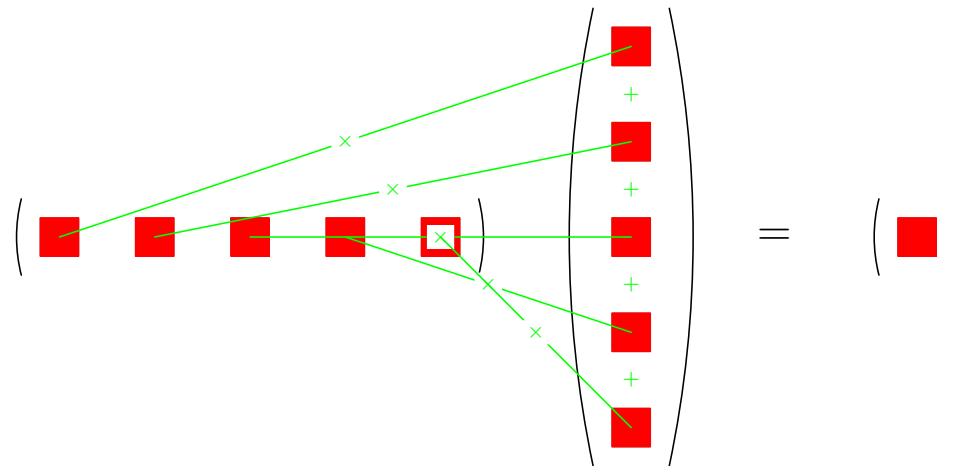
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$$\begin{pmatrix} \blacksquare & \blacksquare & \blacksquare & \blacksquare & \blacksquare \end{pmatrix} \begin{pmatrix} \blacksquare \\ \blacksquare \\ \blacksquare \\ \blacksquare \\ \blacksquare \end{pmatrix} = \begin{pmatrix} \blacksquare \end{pmatrix}$$

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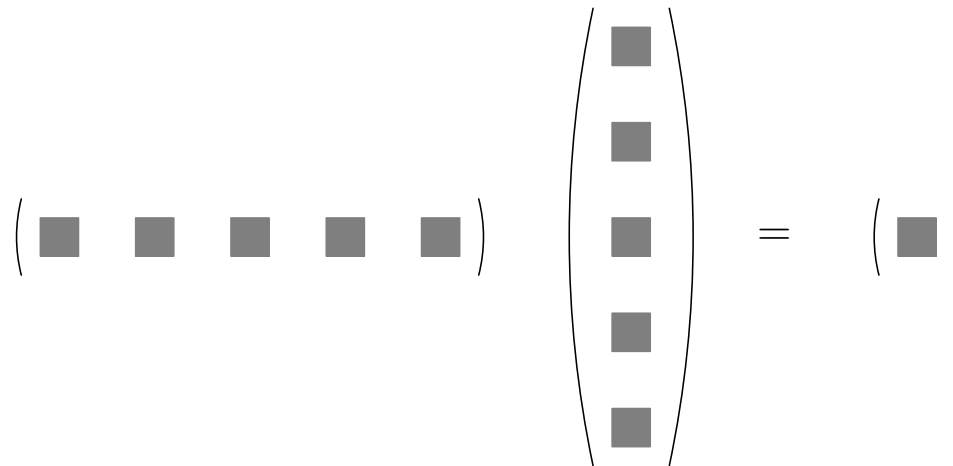
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$$\left(\begin{array}{ccccc} \blacksquare & \blacksquare & \blacksquare & \blacksquare & \blacksquare \end{array} \right) \left(\begin{array}{c} \blacksquare \\ \blacksquare \\ \blacksquare \\ \blacksquare \\ \blacksquare \end{array} \right) = \left(\blacksquare \right)$$

- Inner products encode angles between vectors

$$\mathbf{a}^T \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta)$$

Outer Products

- We can also define the outer-product between any size vectors,
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Outer Products

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The diagram illustrates the outer product operation ab^T . On the left, a column vector a of size 4x1 is represented by a large left parenthesis followed by four gray squares stacked vertically, and a large right parenthesis. This is multiplied by a row vector b^T of size 1x3, represented by a large left parenthesis followed by three gray squares, and a large right parenthesis. An equals sign follows, leading to the resulting 4x3 matrix. This matrix is shown as a large left parenthesis followed by four rows, each containing three gray squares, and a large right parenthesis.

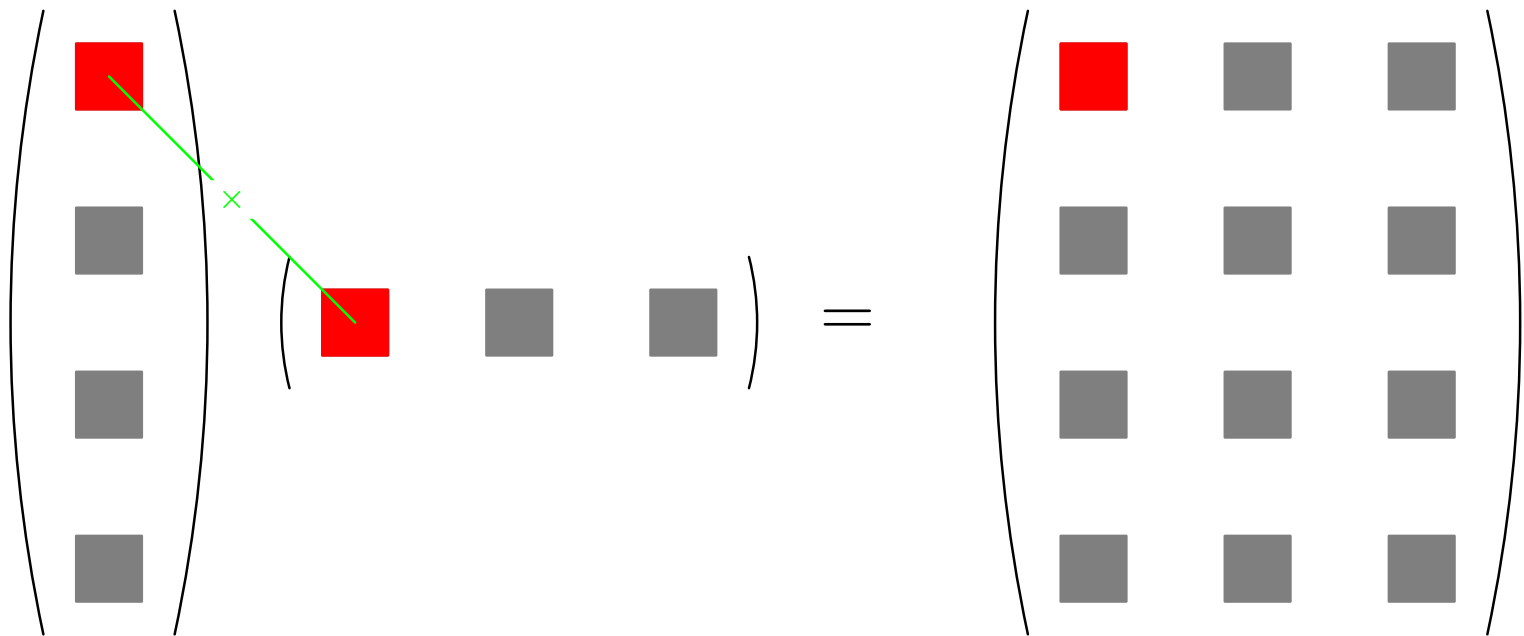
Outer Products

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The diagram illustrates the outer product operation ab^T . On the left, a column vector a of size 4x1 is shown, with its top element highlighted in red. This is multiplied by a row vector b^T of size 1x3, also with its first element highlighted in red. The result is a 4x3 matrix where the first column contains the original elements of a (one red, three gray), and the remaining two columns are gray squares, representing the scaling of a by the other elements of b .

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The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4x1 is shown, with its top element highlighted in red. This is multiplied by a row vector of size 1x3, also with its middle element highlighted in red. The result is a 4x3 matrix where the top row is entirely red, and the other three rows are entirely gray, representing the scaling of the row vector by each element of the column vector.

$$\begin{pmatrix} \text{red} \\ \text{gray} \\ \text{gray} \\ \text{gray} \end{pmatrix} \begin{pmatrix} \text{gray} & \text{red} & \text{gray} \end{pmatrix} = \begin{pmatrix} \text{red} & \text{red} & \text{red} \\ \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \end{pmatrix}$$

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The diagram illustrates the outer product of two vectors. On the left, a column vector is shown with four elements: a red square at the top, followed by three gray squares. To its right is a row vector with three elements: two gray squares followed by a red square. An equals sign follows, and to the right is a 4x3 matrix. The matrix is formed by multiplying each element of the column vector by each element of the row vector. The top row of the matrix has a red square in the third column, and all other elements are gray squares.

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The diagram illustrates the outer product operation ab^T . On the left, a column vector a of size 4x1 is shown, with its second element highlighted in red. This is multiplied by a row vector b^T of size 1x3, with its first element highlighted in red. The result is a 4x3 matrix where each row is a scalar multiple of the vector b^T , and the first column is a scalar multiple of the vector a . The red squares indicate the element-wise multiplication that defines each entry in the resulting matrix.

Outer Products

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The diagram illustrates the outer product of two vectors. On the left, a column vector a is shown with four elements: three gray squares and one red square in the second position. This is multiplied by a row vector b^T with three elements: two gray squares and one red square in the middle position. The result is a 4x3 matrix where each element is the product of the corresponding elements from a and b^T . The resulting matrix has four rows and three columns. The second row and second column element is red, while all other elements are gray.

Outer Products

- We can also define the outer-product between any size vectors, ab^T

The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4 is shown, with its second element highlighted in red. This is multiplied by a row vector of size 3, with its third element highlighted in red. The result is a 4x3 matrix where the third column is red and all other elements are gray.

$$\begin{pmatrix} \text{gray} \\ \text{red} \\ \text{gray} \\ \text{gray} \end{pmatrix} \begin{pmatrix} \text{gray} & \text{gray} & \text{red} \end{pmatrix} = \begin{pmatrix} \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{red} \\ \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \end{pmatrix}$$

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The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4 is shown, with its third element highlighted in red. This is multiplied by a row vector of size 3, with its first element highlighted in red. The result is a 4x3 matrix where every element in the third row is red, and all other elements are gray, representing the outer product ab^T .

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The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4 is shown, with its third element highlighted in red. This is multiplied by a row vector of size 3, with its third element also highlighted in red. The result is a 4x3 matrix where every element in the third row and third column is red, while all other elements are gray.

$$\begin{pmatrix} \square \\ \square \\ \color{red}\square \\ \square \end{pmatrix} \begin{pmatrix} \square & \square & \color{red}\square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \color{red}\square & \color{red}\square & \color{red}\square \\ \square & \square & \square \end{pmatrix}$$

Outer Products

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The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4 is shown, with the bottom element highlighted in red. This is multiplied by a row vector of size 3, with the first element highlighted in red. The result is a 4x3 matrix where each row is a scaled version of the row vector, and the first column is a scaled version of the column vector. The bottom-left element of the resulting matrix is red, representing the product of the two highlighted elements.

$$\begin{pmatrix} \square \\ \square \\ \square \\ \color{red}\square \end{pmatrix} \begin{pmatrix} \color{red}\square & \square & \square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \\ \color{red}\square & \square & \square \end{pmatrix}$$

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The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4 is shown, with the bottom element highlighted in red. This is multiplied by a row vector of size 3, also with the middle element highlighted in red. The result is a 4x3 matrix where the bottom row has its middle element highlighted in red, representing the outer product of the original vectors.

$$\begin{pmatrix} \square \\ \square \\ \square \\ \color{red}\square \end{pmatrix} \begin{pmatrix} \square & \color{red}\square & \square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \\ \square & \color{red}\square & \square \end{pmatrix}$$

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$$\begin{pmatrix} \square \\ \square \\ \square \\ \color{red}\square \end{pmatrix} \begin{pmatrix} \square & \square & \color{red}\square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \\ \color{red}\square & \color{red}\square & \color{red}\square \end{pmatrix}$$

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The diagram illustrates the outer product of a column vector and a row vector. On the left, a column vector is shown with four elements: three gray squares and one red square at the bottom. This is multiplied by a row vector with three elements: two gray squares and one red square on the right. The result is a 4x3 matrix on the right, where each element is the product of the corresponding elements from the input vectors. The bottom-right element is red, while all other elements are gray.

$$\begin{pmatrix} \square \\ \square \\ \square \\ \color{red}\square \end{pmatrix} \begin{pmatrix} \square & \square & \color{red}\square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \\ \square & \square & \color{red}\square \end{pmatrix}$$

- Notice that we need to know whether we are dealing with row or column vectors

Norms

- Norms are non-negative numbers associated with the “size” of a vector
- The norm you should be familiar with is the **Euclidean Norm** or 2-norm

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

- We will see there exist other norms (using different ways of measuring size)
- The Euclidean norm is intimately related to the inner-product, we will see other examples of this for different vector spaces

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Matrices as Linear Mappings

- Matrices represent mappings between vectors

$$\mathbf{b} = \mathbf{M}\mathbf{a}$$

- The elements of $\mathbf{b}$ are **linear sums** of the elements of $\mathbf{a}$

$$b_i = \sum_j M_{ij} a_j$$

- Two important consequences of this linearity are for any real c

$$\mathbf{M}(c\mathbf{a}) = c(\mathbf{M}\mathbf{a})$$

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- Matrices represent mappings between vectors

$$\mathbf{b} = \mathbf{M}\mathbf{a}$$

- The elements of $\mathbf{b}$ are **linear sums** of the elements of $\mathbf{a}$

$$b_i = \sum_j M_{ij}a_j = M_{i1}a_1 + M_{i2}a_2 + \cdots + M_{in}a_n$$

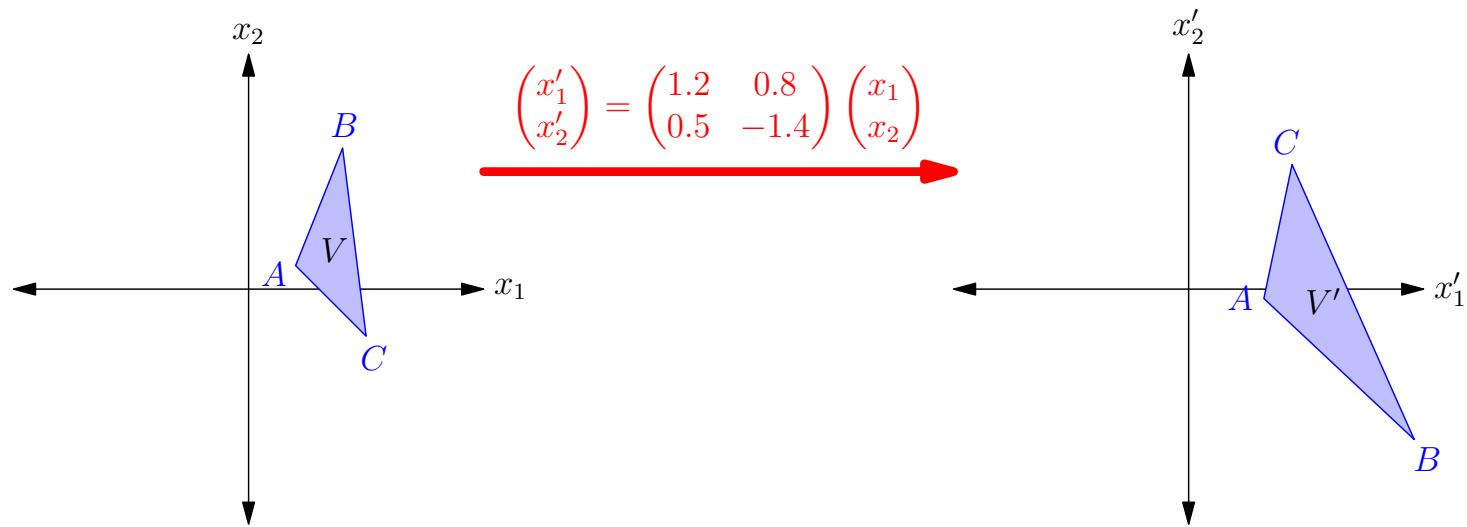
- Two important consequences of this linearity are for any real c

$$\mathbf{M}(c\mathbf{a}) = c(\mathbf{M}\mathbf{a})$$

$$\mathbf{M}(\mathbf{a} + \mathbf{b}) = (\mathbf{M}\mathbf{a}) + (\mathbf{M}\mathbf{b})$$

Action Of Matrices on Volumes

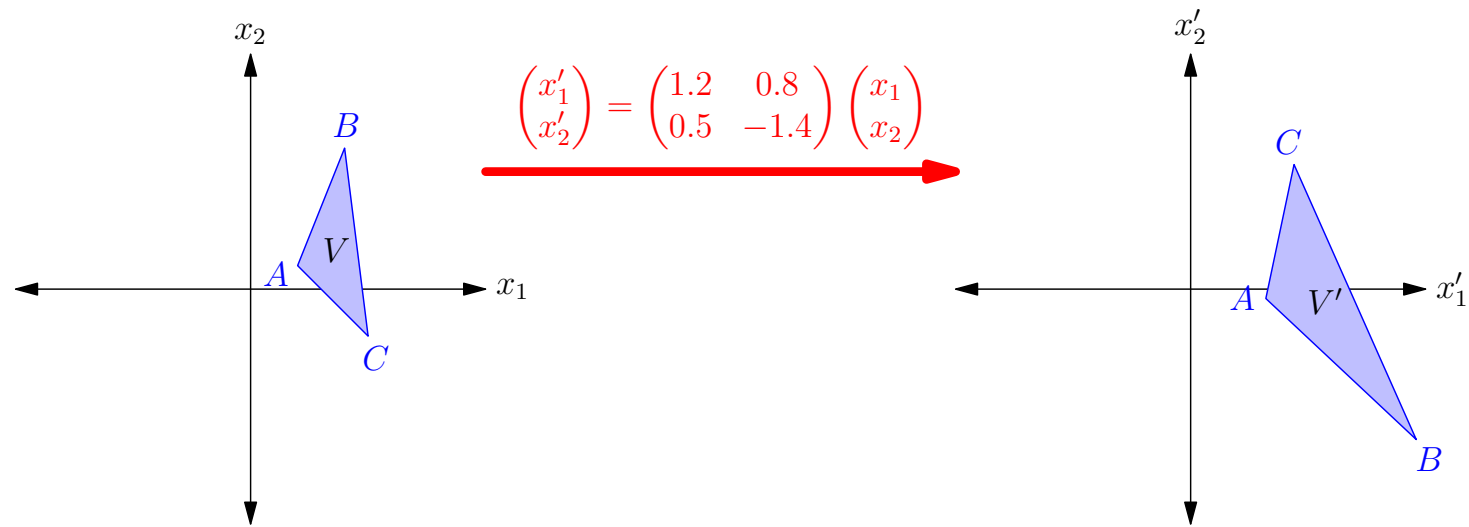
- For a closed set of points corresponding to a volume, V in $\mathbb{R}^n$, treating each point as a vector, we can ask what happens to the volume after mapping the vector by a matrix



- For square matrices, $\mathbf{M}$, the determinant is equal to $\det(\mathbf{M}) = \pm V'/V$
- The sign tells us if there has been a reflection

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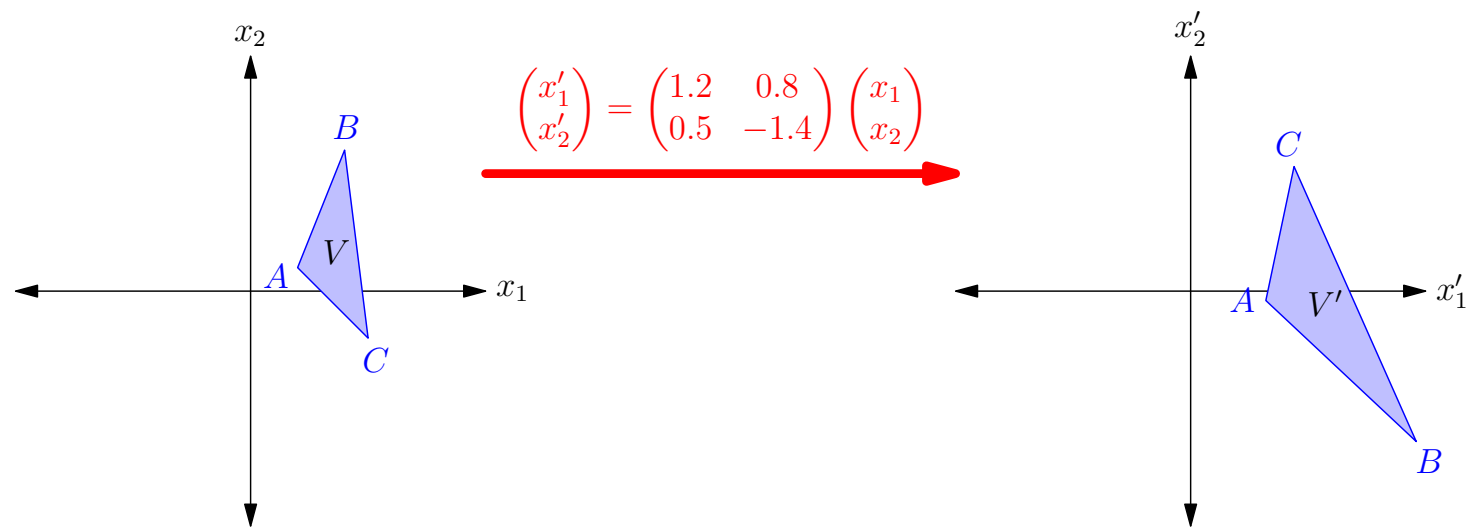
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Determinants

- A consequence of the fact that $\mathbf{AB}$ denotes the composition, apply mapping $\mathbf{B}$ then apply mapping $\mathbf{A}$

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \times \det(\mathbf{B})$$

- Determinants are only defined for square matrices, as non-square matrices project into different size spaces, where the volumes cannot be compared
- **Singular matrices** projects all points onto a lower dimensional space, so have determinant 0

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Multivariate Functions

- In machine learning we heavily use multivariate functions

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

- Often our functions will be scalar value $f : \mathbb{R}^n \rightarrow \mathbb{R}$
- E.g. a learning machine for regression might be $\hat{f}(\mathbf{x}, \mathbf{w})$
- We say a function f is linear if

$$f(c\mathbf{x}) = cf(\mathbf{x}) \qquad f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$$

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Gradients

- The gradient of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ at a point $\mathbf{x}^*$ is defined as the column vector

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f(\mathbf{x}^*)}{\partial x_1} \\ \frac{\partial f(\mathbf{x}^*)}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x})^*}{\partial x_n} \end{pmatrix}$$

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- To reason about events with uncertain outcomes we introduce **random variables**
- Random variables corresponds to objects that we assign numbers to depending on the outcome of a random event
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- A continuous random variable might be the temperature at 12 noon tomorrow in Hyde park

Probability Distributions

- For discrete events, we can define a **probability mass function** $\mathbb{P}(X = x)$ that the random variable takes the value x
- This is a number between 0 and 1 with $\sum_{x \in \mathcal{X}} \mathbb{P}(X = x) = 1$
- For continuous events the probability of X taking a particular value x is zero
- Instead we consider the probability density $f_X(x)$ which we can define as

$$f_X(x) = \lim_{\delta x \rightarrow 0} \frac{\mathbb{P}(x < X \leq x + \delta x)}{\delta x}$$

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Expectations

- One of the most useful quantities in dealing with random numbers is to look at the average value, aka as the **expectation**, for some function $g(X)$ of the random variable

$$\mathbb{E}[g(X)] = \begin{cases} \sum_{x \in \mathcal{X}} \mathbb{P}(X = x) g(x) \\ \int f_X(x) g(x) dx \end{cases}$$

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$$\mathbb{E}[cg(X)] = c\mathbb{E}[g(x)]$$

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- We will revisit most of these topics to gain a better understanding, but you will be lost in the other machine learning modules if you are not familiar with everything I've covered
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